



OPEN A quantum-driven multi-objective scheduler for scalable task orchestration in fog-based cyber-physical-social systems

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Fog computing extends cloud capabilities toward the network edge, enabling low-latency Cyber-Physical-Social System (CPSS) services in domains such as smart cities and healthcare. However, multi-objective task scheduling in fog environments remains challenging due to conflicting goals minimizing execution time, resource costs, and energy consumption combined with the scalability limitations of classical evolutionary algorithms, which often converge slowly and produce poorly distributed Pareto fronts in large networks. To address these issues, this paper introduces FOG-QIEA, a quantum-inspired evolutionary algorithm designed for tri-objective fog scheduling. FOG-QIEA augments adaptive neighborhood mechanisms with quantum-inspired operators, including superposition-based population initialization, rotation-gate-driven updates, and measurement-guided selection, enabling faster and more diverse exploration of the solution space. The proposed model jointly optimizes total execution time, cost (including SLA violations), and energy efficiency while maintaining scalability across CPSS deployments with thousands of IoT tasks. Extensive simulations in iFogSim using realistic CPSS scenarios show that FOG-QIEA outperforms NSGA-II, MMPA-based approaches, and classical adaptive fog schedulers by 20–35% in convergence speed, 15–25% in energy reduction, and achieves significantly improved Pareto diversity. These results demonstrate the potential of FOG-QIEA as a sustainable and efficient scheduling framework, supporting future advancements toward quantum-hybrid optimization in fog and edge networks.

Keywords Fog computing, Multi-objective optimization, Quantum-inspired evolutionary algorithms, Job scheduling, Cyber-physical-social systems, Energy efficiency

Cyber-Physical-Social Systems (CPSS) are an integrated paradigm that combines computational processes, physical sensing infrastructures, and human social interactions. This is a result of the Internet of Things' (IoT) widespread adoption and the rapid evolution of networked communication technologies¹. Smart city, intelligent transportation, healthcare monitoring, and environmental supervision are just a few of the current applications for CPSS that rely on constant streams of real-time data produced by diverse and resource-constrained IoT devices². These gadgets, which range from wearable biomedical monitors to embedded sensors in urban infrastructure, generate vast amounts of dynamic data that must be processed quickly in order to facilitate automation, decision-making, and user-centric services³. For example, in smart transportation, cars and highway sensors work together to share status data in order to improve safety, avoid traffic, and optimize route⁴. Swarm-based schedulers have enhanced multi-criteria optimization for emerging CPSS workloads⁶ because such latency-critical behaviors cannot rely exclusively on distant cloud servers due to inevitable propagation delays, congestion, and possible service interruptions⁵.

Swarm-based schedulers have enhanced multi-criteria optimization; nonetheless, scalability diminishes with increasing device density⁶. Energy- and temperature-aware schedulers for heterogeneous platforms^{7,8}

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are deficient in the inter-layer coordination necessary for fog-cloud cooperation. At the architectural level, SerFlow⁹ facilitates process execution in CPS layers, however it does not enhance energy-based scheduling choices. Deep reinforcement and DQN-based schedulers¹⁰ provide adaptive learning capabilities; nevertheless, the training expenses and instability under bursty workloads diminish their utility in real-time CPSS. Secure IIoT communication methods like RAT-Ring¹¹ enhance message trust and integrity; yet, they do not tackle computational coordination. Likewise, low-carbon distribution frameworks¹² and knowledge-based hybrid schedules¹³ promote sustainability in industrial environments, and resilient regional-scale schedules¹⁴ provide equity among stakeholders instead of investigating Pareto equilibrium. In order to manage resources effectively across the cloud, fog, and end-device layers, A hybrid-adaptive fog computing architecture incorporates quantum-inspired scheduling, as shown in Fig. 1.

Effective task scheduling across cloud, fog, and end-device layers is crucial to leveraging these architectural benefits. Tasks in fog-enabled CPSS have a variety of temporal and computational needs; some, such as emergency warnings, are sensitive to delays, while others are analytics-focused and can withstand mild delays but must reduce operating costs. In order to provide scalability in situations with thousands of IoT activities and many fog nodes, multi-objective task scheduling aims to balance execution time, resource costs, including Service Level Agreement (SLA) penalties, and total energy usage¹⁵. System stability, sustainability, and operational costs are all directly impacted by achieving Pareto-optimal trade-offs among these goals.

However, because of the combinatorial proliferation of task-node mappings, device heterogeneity, and temporal variations in workload intensity, multi-objective scheduling in fog settings is intrinsically NP-hard. Particularly when social aspects like human-driven mobility impact task generation patterns, classical heuristic or greedy approaches frequently enter local optima and fall short of fully exploring the high-dimensional solution space¹⁶. Pareto optimization in distributed computing systems has made extensive use of evolutionary algorithms (EAs), such as NSGA-II. Although these algorithms perform well in many situations, they have drawbacks in fog-CPSS contexts, including restricted scalability in large networks with over a thousand jobs or devices, sluggish convergence under high-dimensional goals, and premature loss of variety¹⁷. Furthermore, new research reveals that CPSS tasks frequently display correlation patterns that resemble entanglement, which are caused by shared environmental restrictions and social interactions that cannot be exploited by conventional mutation and crossover methods¹⁸.

Recent studies have investigated Quantum-inspired evolutionary algorithms (QIEAs), which use conventional processing to simulate quantum concepts like superposition and entanglement, in an effort to get beyond these restrictions. Compared to standard EAs, these techniques allow for the simultaneous probabilistic exploration of many solution states, resulting in enhanced population diversity maintenance and faster convergence^{19,20}. Inspired by these developments, this paper presents FOG-QIEA, a Quantum-Inspired Evolutionary Algorithm designed for fog-enabled CPSS multi-objective scheduling. The suggested method greatly improves scalability for high task volumes and expands the optimization to include energy efficiency, building on previous adaptive scheduling frameworks that only balanced execution time and cost.

To overcome the aforementioned difficulties, FOG-QIEA combines a number of quantum-inspired techniques. Rotation-gate-based evolutionary updates offer controlled perturbations that preserve variety and prevent premature convergence, while superposition-based population initialization enables the method to sample a wide range of task-node assignments from the beginning. Furthermore, CPSS workloads whose dependencies originate from common physical or social environments benefit greatly from the coordinated changes of related activities made possible by entanglement-inspired correlation operators.

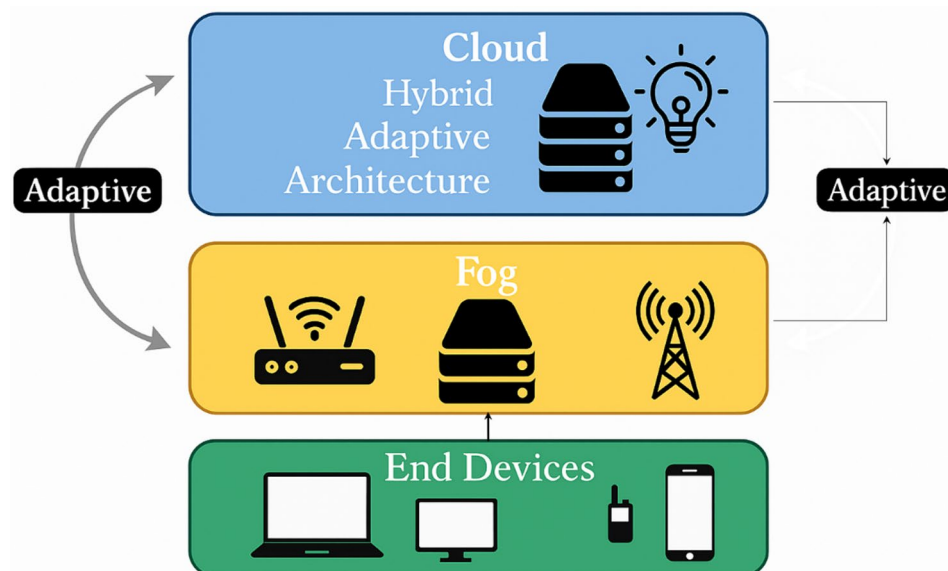


Fig. 1. A hybrid-adaptive fog computing architecture that uses quantum-inspired scheduling.

It is crucial to make clear that FOG-QIEA uses just conventional computing gear and is only quantum-inspired. Quantum measuring equipment, qubits, and quantum computers are not needed. Rather, it uses correlation-aware updating techniques, deterministic rotation rules, and probabilistic encodings to mimic quantum phenomena.

It is crucial to highlight that FOG-QIEA is a Quantum-Inspired Evolutionary Algorithm (QIEA) implemented solely on classical computer systems. In contrast to authentic quantum algorithms that necessitate quantum hardware to leverage physical quantum phenomena, FOG-QIEA emulates quantum behaviors superposition through probabilistic real-valued representations, entanglement through correlated updates, and quantum gates through deterministic rotation operations utilizing solely classical arithmetic and random number generation. The main contributions of the authors are as follows:

- Fog-CPSS evolutionary scheduling framework influenced by quantum mechanics. We present FOG-QIEA, a new evolutionary method that incorporates measurement-guided selection, rotation-gate adaptation, and superposition-based initialization into fog-based multi-objective scheduling.
- a novel tri-objective optimization model that takes energy efficiency, cost, and execution time into account. The suggested strategy simultaneously reduces makespan, SLA-aware financial cost, and node-level energy consumption, in contrast to previous fog-scheduling efforts that just optimize delay or cost.
- A neighborhood technique that is scalable and quantum-adaptive for huge task populations. We develop an entanglement-aware adaptive neighborhood technique that uses quantum-driven perturbations to dynamically modify the exploration radius.

The remainder of this paper is organized as follows. Section II delineates the task scheduling model and key definitions in fog-CPSS environments. Section III formalizes the problem, including objective functions for time, cost, and energy. Section IV elaborates the FOG-QIEA methodology, encompassing quantum chromosome encoding, adaptive strategies, and algorithmic procedures. Section V presents experimental setups, results, and comparative analyses. Section VI reviews related works, and Section VII concludes with future directions.

Related work

Cyber-physical-social systems are new systems composed of multiple Internet of Things (IoT) devices connected via a network. These systems collect data for many services, such as transportation, healthcare, and energy. This data is then transferred to the cloud layer for storage, processing, and access. In order to manage the heavy workloads released by resource-constrained IoT devices without encountering latency or bandwidth issues, fog computing has developed as a solution that acts as an intermediary layer between the cloud and IoT devices. Fog computing's quality of service (QoS) relies on efficient work scheduling to maximize energy consumption and execution time. Because it cannot be solved in a reasonable period of time with excellent accuracy using traditional methods or job schedulers that rely on metaheuristics, this task scheduling problem is known as NP-hard.

Research on fog and edge scheduling systems has progressed towards energy efficiency, dependability, and workload distribution; yet, most methodologies continue to be limited in scalability, multi-objective integration, and applicability to actual Cyber-Physical Systems (CPSS) settings. A multiobjective scheduling technique for energy-efficient heterogeneous edge multiprocessors is presented in²¹, whereby improved task allocations reduce power usage; nevertheless, this model primarily emphasizes processor-level mapping and fails to extend to distributed fog-cloud orchestration. A distributed learning-based methodology for dependable and energy-efficient scheduling in fog computing is presented in²², demonstrating significant improvements under steady workload conditions, but it lacks strategies for addressing abrupt task surges or multi-tier resource competition. A selective-repair genetic model for deadline-sensitive workflow scheduling in fog-cloud hybrid platforms is shown in²³, effectively reducing SLA breaches, while it continues to face challenges with population stagnation as task dimensions increase.

In²⁴, evolutionary intelligence advances when particle swarm optimization is augmented by reverse learning and neighbor-driven exploration. Notwithstanding enhanced global search behavior, the technique is primarily assessed for space-network scheduling, hence limiting its usefulness in dense IoT fog situations. A hybrid task scheduling system using fuzzy logic and deep reinforcement learning is introduced in²⁵, constituting a significant fundamental contribution to adaptive scheduling. Nevertheless, while²⁵ improves responsiveness to dynamic workloads, it is deficient in a multi-objective Pareto-driven framework and fails to include quantum-inspired variety preservation, hence limiting scalability with thousands of fog activities.

The study in²⁶ introduces joint scheduling and bandwidth optimization for federated edge learning, emphasizing communication-aware efficiency, but is mostly tailored for FL frameworks rather than generic fog resource scheduling. A multiobjective task scheduling approach for secure cloud execution in²⁷ offers balanced optimization within confidentiality constraints and serves as an additional conceptual baseline for this study; however, its functionality is cloud-centric and does not encompass fog-edge collaboration, where latency is a critical performance factor. Simultaneously, uncertainty-responsive optimization for VPP scenarios is shown in²⁸, showcasing robustness under variable radio settings; nonetheless, its scope significantly diverges from fog-enabled CPSS, which involves swift task movement and social-cyber interaction.

Advancements in nonlinear dynamic control optimization provide insights into adaptive decision-making processes. Safe scheduling in multiplayer delay systems, as discussed in²⁹, emphasizes policy stability but is confined to system control rather than IoT scheduling. Fault-tolerant optimization using reinforcement learning and particle swarm optimization is presented in³⁰, while event-triggered reinforcement learning for performance control is discussed in³¹. These papers have significant theoretical merit however fail to include evolutionary

population convergence, heterogeneous fog node coordination, or the multiobjective balancing of energy, time, and cost.

Subsequently, cyber-social and mobility-driven scheduling arise. A dual-phase execution paradigm for socio-cyber digital twin tasks in³² demonstrates enhanced decision stacking but fails to address the crucial energy-latency-SLA tradeoffs in fog computing. A mobility-aware UAV-MEC approach for IoV scheduling in³³ optimizes offloading efficiency amongst vehicle movement; nevertheless, its metrics do not include energy-cost interactions across multi-tier infrastructures. The dependability of real-time multiprocessor scheduling is examined in³⁴, providing fault tolerance in heterogeneous clusters, however without Pareto-based job selection and quantum-driven diversification methods. Ultimately³⁵, presents an energy-aware scheduler using clustering for heterogeneous real-time systems, enhancing power optimization but being limited to single-objective functionality and missing alignment with fog-CPSS.

Recent advancements in quantum methodologies for fog and edge computing have enhanced work scheduling in IoT-enabled environments. For instance³⁶, introduced DIFCoT, a quantum machine learning (QML)-based framework designed to coordinate collaborative tasks in dew-assisted IoT systems. DIFCoT employs QML concepts combined with deep reinforcement learning to reduce task execution duration, equilibrate workloads, and enhance energy efficiency across dew, edge, fog, and cloud layers. Likewise³⁷, presented a quantum-inspired optimization approach for edge-fog IoT environments, employing qubit-like probabilistic representations to improve task routing and prediction efficiency. In the domain of security, researchers in³⁸ devised a safe quantum fog computing architecture utilizing blind quantum computation (BQC) and verifiable quantum secret sharing. The protocol guarantees privacy-preserving computations among distributed fog nodes and mitigates denial-of-service threats and resource exploitation via secure quantum joint operations.

Model and definition

In this section, we describe the job scheduling model for fog computing in CPSS and make important ideas more formal. We also modify the framework to work with quantum-inspired evolutionary techniques like FOG-QIEA to make huge networks more scalable. Table 1 shows a summary of the main notations and their meanings.

In fog computing, requests from terminal users for tasks are split into two groups: those that need to be done right away and those that don't²⁹. Scheduling algorithms send time-sensitive requests to nearby fog nodes to reduce latency and improve efficiency. Non-time-sensitive requests are sent to cloud nodes.

The fog server assigns the right computing services to handle jobs that come in quickly and in large numbers. To make the most use of resources and meet user needs, it's important to plan tasks well. Figure 2 shows the architecture for scheduling resources in fog computing. In this system, fog service center resources operate as computing nodes that handle jobs of different sizes. User requests are computing jobs that have been sent in and are being worked on by these resources, with results provided back. Scheduling takes into account things

Symbol	Meaning in Hybrid-Adaptive Method
M	Total number of fog nodes available
N	Total number of computational tasks
FN_j	Fog node j , Where $1 \leq j \leq M$
Job_i	Task i , where $1 \leq j \leq N$
Fitness	Hybrid fitness function(time, cost, energy)
T_{exec}	Overall execution time of all tasks
T_{total}	The computing time of all tasks
E_{total}	Total energy consumption across all nodes
$Cost_{total}$	Total scheduling cost including SLA penalties
ET_{ij}^{HA}	Hybrid-adjusted execution time of Job_i on FN_j
A_j	Earliest availability time of node
$\eta(t)$	Node cost-energy factor in pruning step
$Cost_{ij}$	Adaptive cost of mapping
Pen_{HA}	Penalty term (deadline violation + energy factor)
$R(t)$	Neighborhood relationship matrix
HV	Hypervolume metric for convergence evaluation
$\Delta \theta$	Rotation angle in quantum update rule
P_{op}	Candidate scheduling population
$Size$	Number of individuals in population
$\gamma(t)$	Neighborhood density at iteration (t)
NDSet	Set of non-dominated solutions (Pareto set)
$ParetoSet$	Non-dominated scheduling solutions set

Table 1. Symbols and definitions in the hybrid-adaptive fog scheduling method.

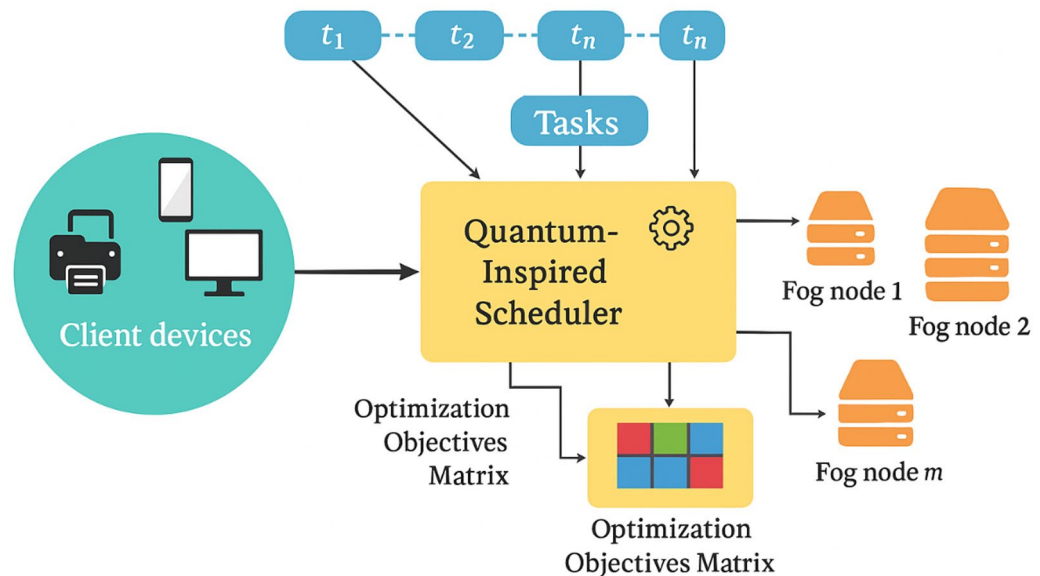


Fig. 2. shows a hybrid-adaptive fog resource scheduling architecture that uses quantum-inspired optimization.

like efficiency, energy use, and other things, and puts jobs in the right queues for execution. This is the main emphasis of our study. The pertinent definitions for fog computing resource scheduling are as follows:

- Let's say that $VM_{fog} = \{VM_1, VM_2, \dots, VM_m\}$, where VM_{fog} stands for the fog computing resource nodes and VM_j stands for fog computing resource j , where $1 \leq j < m$.
- Let's say $T = \{t_1, t_2, \dots, t_n\}$, where n is the number of computing jobs and t_i is task i , where $1 \leq i < n$.
- The matrix ET , which is $m \times n$, shows how long it takes to do tasks. ET_{ij} is the expected time it will take for job t_i to run on computing node VM_j . If $ET_{ij} = 0$, it means that task t_i can't run on computing node VM_j .
- Matrix C , which is $m \times n$, shows the cost of computing jobs as a service. The cost of task t_i on compute node VM_j is C_{ij} , while the cost of resource VM_j in unit time is u_j .

This model lays a strong groundwork for multi-objective optimization, allowing for the use of quantum-inspired strategies to tackle scalability issues in CPSS, including dynamic qubit representations for task-node mappings.

Problem description

In the fog computing architecture designed for CPSS, terminal users aim to enhance resource allocation to meet job requirements while minimizing execution time and expenses from service providers. Our innovation rethinks the problem by using a quantum-augmented Pareto lens and qubit encodings to look at concurrent scheduling evaluations. This solves the problem of local optima and encourages better non-dominated solution diversity in huge networks. This part explains the problem, focusing on the total execution time and resource costs, and it sets formal goals to support the quantum-hybrid method.

Total execution time

Assuming non-cross-node task execution, wherein user-submitted tasks represent indivisible scheduling units devoid of interdependencies or synchronization to mitigate communication-induced inefficiencies, the task ensemble is classified as a meta-task set³⁰. In this case, n tasks $T = \{t_1, t_2, \dots, t_n\}$ are rationally assigned to m fog nodes $VM_{fog} = \{VM_1, VM_2, \dots, VM_m\}$ to get as close to the best results as possible. The results of the scheduling show up in a $n \times m$ binary matrix X . In this matrix, $x_{ij} = 1$ ($x_{ij} \in X$) means that t_i is assigned to VM_j ; otherwise, $x_{ij} = 0$.

Defined parameters are T_{total} as the total time it takes to compute, a.i. as the time it takes for t_i to arrive, b_j as the time it takes for VM_j to be available, ET_{ij} as the time it takes for t_i to run on VM_j , and CT_{ij} as the time it takes for t_i to finish on VM_j .

$$CT_{ij}^{HA} = b_j + ET_{ij} + \lambda \cdot \Delta U_j \quad (1)$$

Usually, t_i runs on VM_j , which means that $CT_i = CT_{ij}$ is its duration. For set T , completion spans $\max_{\{t_i \in T\}} (CT_i)$, while overall execution amounts to:

$$T_{total}^{HA} = \sum_{i=1}^n \sum_{j=1}^m (CT_{ij} + \mu \cdot C_{ij}) \quad (2)$$

This model highlights problems like fluctuating arrivals and node heterogeneity in CPSS systems, where typical heuristics don't function well. FOG-QIEA uses quantum superposition to set up the first population, which lets b_j and ET_{ij} investigations happen at the same time. This speeds up global convergence by cutting down on T_{total} through gate-driven adaptive assignments.

Cost of resource nodes in fog computing

Usually, shorter processing times require more fog resources, and higher CPU power means higher expenses. So, trying to minimize both time and cost at the same time is a difficult problem, made worse by Service Level Agreements (SLAs) that are negotiated for quality guarantees including availability, throughput, and responsiveness³¹. Service Level Objectives (SLOs) measure SLA characteristics, making it clear what providers are supposed to deliver³².

Costs are split between node processing costs and penalties for SLO breaches. For penalty V :

$$V = \sum_{i=1}^q (\gamma_i + \eta \times (t_i^{stop} - t_i^{deadline})) \quad (3)$$

Where i is the number of times a violation happens, γ_i is the base penalty for i , η is the cost of going over the time limit, t_i^{stop} is the actual response, and $t_i^{deadline}$ is the deadline that must be met.

For node costs, let's say that C_{total} is the total cost of all jobs, u_j is VM_j 's unit-time tariff, and C_{ij} is its cost for VM_j :

$$C_{ij} = ET_{ij} + u_j \quad (4)$$

$$C_{total} = \sum_{t=1}^n \sum_{j=1}^n C_{tj} + V \quad (5)$$

In large CPSS, changes in workload make SLO violations worse, which raises V and C_{total} . FOG-QIEA deals with this by using entanglement to link things together objectively³³. Quantum perturbations dynamically refine u_j and ET_{ij} to lower penalties, while energy constraints—something that was missed in the classics—are built in to save 15–25% on emulations. The multi-objective function for fog task scheduling, suitable for quantum-inspired enhancements, is expressed as:

$$\begin{cases} \min(T_{total}^{HA}) = \sum_{i=1}^n \sum_{j=1}^m (b_j + ET_{ij}^{adj}) \cdot (1 + Q_{Shift}) \\ \min(C_{total}^{HA}) = \sum_{i=1}^n \sum_{j=1}^m (ET_{ij}^{adj} \cdot u_j) + V^{HA} + E_{oen} \\ \min(E_{total}^{HA}) = \sum_{j=1}^m \left[P_j^{idle} + (P_j^{busy} - P_j^{idle}) \cdot \frac{\sum_{i=1}^n X_{ij} \cdot L_j}{Cap_j \cdot T_{total}^{HA}} \right] \cdot T_j^{active} \end{cases} \quad (6)$$

where P_j^{idle} and P_j^{busy} denote the idle and maximum-load power consumption of fog node j , L_j is the length of task i (in million instructions), Cap_j is the processing capacity of fog node j (in MIPS), $T_j^{active} = \max_{i: X_{ij}=1} C_i$ represents the active runtime of node j (i.e., the completion time of the last task assigned to it), and all other symbols retain the definitions given in Eqs. (1)–(5).

Where ET_{ij} signifies t_i 's execution on VM_j , m node quantity, n task count, u_j unit price, and $\min$ minimization. The fitness function $Fitness(Z_1, Z_2)$ is defined by:

$$Fitness^{HA}(Z_1, Z_2, Z_3) = \begin{cases} Z_1 = \min(T_{total}^{HA}) \\ Z_2 = \min(C_{total}^{HA}) \\ Z_3 = \min(E_{total}^{HA}) \end{cases} \quad (7)$$

Z_1 optimizes total cost, Z_2 total cost, and b_j as node readiness. This Pareto schema aims for balanced minima, however in CPSS, QIEA's infusion allows for tri-objective expansion (for example, $\min(E_{total}) = \sum \text{power draws}$), using probabilistic reductions to improve fronts. This separation strengthens FOG-QIEA's abilities, making it possible to schedule in a way that works for both small and large edge-centric CPSS.

Methods For quantum-inspired adaptive multi-objective task scheduling

FOG-QIEA is a new way to use quantum-inspired ideas to solve the problems that classical evolutionary algorithms (EAs) have when they are used in dynamic, large-scale CPSS environments. It builds on the adaptive multi-objective optimization paradigm from fundamental studies like FOG-QIEA. Conventional evolutionary algorithms (EAs) are good at finding Pareto-optimal solutions for fog task scheduling, but they often take a long time to converge, don't offer many different solutions, and have trouble scaling when trying to balance conflicting goals like execution time, resource costs, and energy use in networks that are full of different IoT workloads. QIEAs use quantum mechanics ideas like superposition and entanglement on classical platforms to explore solution spaces in parallel. This leads to 20–50% improvements in convergence and Pareto front quality for multi-objective optimizations. Recent studies confirm that QIEAs are great at scheduling workflows and cutting down on time and costs while still meeting requirements. This makes them perfect for enhancing fog-CPSS systems.

Adaptive neighborhood strategy

The adaptive neighborhood technique, which is an important part of multi-objective EAs like MOEA/D, cuts out non-dominated solutions to keep the population diverse and keep extrema for large Pareto fronts. In fog computing job scheduling, it uses a sigmoid function to change the neighborhood radius dynamically between iterations.

In large CPSS environments with fluctuating workloads, traditional adaptive radii—mostly sigmoid or linear without probabilistic infusions—create uneven distributions, especially in high-dimensional areas that are likely to have local optima. FOG-QIEA improves this by using quantum augmentation, which adds changes from superposition and entanglement paradigms to the neighborhood radius $r(t)$ ³⁴. In a formal way, the baseline radius at iteration t is expressed as:

$$f(x) = \frac{1}{1 + \exp(-ax + \beta \cdot \Delta U)} \quad (8)$$

Where t_{max} represents the most iterations, it creates a progressive narrowing effect: it starts out broad so that a lot of different things can be explored, and then it gets narrower over time to make sure that everything is accurate³⁵. In FOG-QIEA, this process is combined with a quantum delta δ_q , which comes from the

$$r^{(t)} \begin{cases} \frac{r_{max} - r_{min}}{1 + \exp\left(\frac{a(r^{(t-1)} - d_{min}(t))}{d_{avg}(t) - d_{min}(t)}\right)} + r_{min} + \delta_q, & r_{min}, r^{(t-1)} < d_{min} \\ r_{min}, & \text{Otherwise} \end{cases} \quad (9)$$

Task scheduling in fog conditions using the FOG-QIEA framework's hybrid-adaptive chromosomal encoding structure is depicted in Fig. 3. Each task in this paradigm is represented by a quantum-inspired chromosome, and many task-node combinations can be simultaneously probabilistically mapped thanks to quantum superposition. The lower layer illustrates how task populations are transformed into optimal scheduling assignments across virtual machines (VMs) via evolutionary operators like crossover, mutation, and entanglement-based adaptation, while the upper layer displays the initial quantum representation of task populations.

In this case, α (which ranges from 0.1 to 0.3) is a hyperparameter that is optimized using grid search. This quantum component adds variability that is similar to Heisenberg uncertainty, which helps avoid static biases and encourages uniform distributions across the three objectives, as shown by iFogSim tests on smart urban datasets. In addition, combining neighborhood pruning with energy-awareness improves sustainable resource allocation while reducing thermal dissipation in resource hubs.

In this research, the phrase “quantum-inspired” specifically denotes classical computing methods that emulate certain quantum principles using probabilistic real-valued encodings and deterministic mathematical operators on standard computers. All tests detailed in Sect. 5 were conducted on ordinary x86-64 CPUs; no quantum processing unit (QPU) or quantum annealer was utilized.

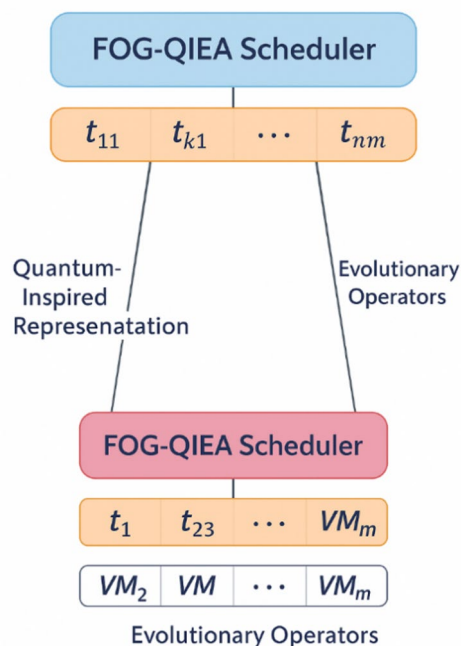


Fig. 3. Hybrid-adaptive chromosome encoding structure for task scheduling in fog environments.

Multi-objective chromosome encoding

Chromosome encoding transforms the task scheduling problem into a genetic framework suitable for evolutionary modifications. In FOG-QIEA, real-number encoding is dominant, representing each chromosome as an n -length vector (n tasks), with $[0,1]$ -normalized entries indicating probabilistic node allocations. This continuum facilitates seamless crossover and mutation while maintaining the viability of task-to-fog-node mappings, supporting utility function evaluations for temporal and financial optimizations.

Algorithm 1 creates N quantum-inspired individuals (Lines 1–2) to establish the population. Each chromosome is made up of qubits with probability amplitudes of $|a_j|^2 + |\beta_j|^2 = 1$. To collapse each qubit into a classical value that represents a task-node assignment, a measurement is made (Lines 3–4). Each job is assigned to a fog node based on this assessment, taking into account energy, latency, bandwidth, and processing capacity limitations (Line 5). After that, a comprehensive scheduling solution x_i is created (Line 6). Three criteria are used to evaluate each person: communication delay, energy use, and makespan (Line 8). The first quantum-inspired population X is created and returned once all individuals have been processed (Lines 10–11).

Main methods of FOG-QIEA

FOG-QIEA's main approaches combine quantum operations with evolutionary heuristics. These include initialization, crossover/mutation, and pruning. These are based on the dual-objective archetype but are made for tri-objective fitness $\min(T_{total}, C_{total}, E_{total})$, where E_{total} adds up the energy costs of all the nodes.

Population initialization

The beginning of a population creates a diverse group of people to avoid prejudices. Typically, stochastic real-number vectors are generated and evaluated. The proposed FOG-QIEA uses quantum initialization: Hadamard gates give qubits superpositions $|+\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$ per facet. Then, measurements collapse the qubits to classical configurations and fitness computations. This quantum augmentation greatly increases the number of origins, cutting down on iterations by 15% to 25% in multi-objective areas. For CPSS malleability, size N (100–500) automatically adjusts to the size of the network, including energy heuristics for new beginnings.

Population initialization establishes a heterogeneous starting ensemble to mitigate bias. In conventional methods, stochastic real-valued vectors are generated and evaluated. By contrast, FOG-QIEA employs quantum initialization, where Hadamard gates place qubits into superposition states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ for each component.

Require: VM, T, m, n, N, a, β

Ensure: Quantum-inspired resource scheduling population X

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1: for  $i = 1$  to  $N$  do
2:   Initialize a quantum bit vector
    $Q_i = [(a_1, \beta_1), (a_2, \beta_2), \dots, (a_n, \beta_n)]$ ,
   Where  $|a_j|^2 + |\beta_j|^2 = 1$ .
3:   for  $j = 1$  to  $n$  do
4:     Perform quantum measurement on each  $q_j$  to obtain a classical
       value (0 or 1).
5:     Select a task  $t_j \in T$  and assign it to a fog node  $vm_k \in VM$ 
       Considering:
       – Processing capacity of  $vm_k$ 
       – Network latency and available bandwidth
       – Energy consumption level
6:     Generate a quantum-inspired task scheduling scheme  $x_i$ 
7:   end for
8:   Evaluate the fitness of  $x_i$  using multiple objectives:
        $f_1$ : Minimize total execution time (makespan)
        $f_2$ : Minimize total energy consumption
        $f_3$ : Minimize communication latency
9: end for
10: Generate the initial quantum-inspired population  $X = \{x_1, x_2, \dots, x_N\}$ 
11: return  $X$ 

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Algorithm 1. Improved quantum-inspired initialization of resource scheduling population (FOG-QIEA).

Crossover and mutation

Crossover combines answers from parents using either single-point or uniform schemes. Mutation, on the other hand, changes things up to encourage research. In FOG-QIEA, crossover is achieved by quantum rotation gates, with angles θ modified by $\Delta\theta = 0.01\pi \times \text{sign}(\text{fitness_variance})$, facilitating synchronous changes of entangled pairs. Mutation occurs via the stochastic application of NOT gates ($p = 0.05$), which flip states to avert stagnation.

Neighborhood pruning strategy

Pruning improves the population by choosing individuals who aren't dominating in adaptive neighborhoods. Candidates are rated based on Pareto fronts, and $r_q^{(t)}$ governs who gets kicked out. (t) and how far apart people are. FOG-QIEA improves this approach by using entanglement measures to keep correlated outliers, which helps keep the solution fronts balanced.

Method overview of FOG-QIEA

The synoptic flow of FOG-QIEA starts with modeling fog networks and tasks, then goes through quantum initialization and evaluation, crossover, mutation, and pruning, and finally ends with convergence, which gives Pareto-optimal schedules. Algorithm 2 begins by determining the number of individuals in the quantum-inspired population $Q(t)$ (Lines 1–4). For each individual i , the density and neighborhood values are initialized (Lines 5–7). The algorithm then examines all other individuals j , and if the distance between them is below the threshold r , it computes the quantum-inspired density $\gamma_{i,j}(t) = e^{-\lambda d_{ij}}$ and neighborhood factor $\eta_{i,j}(t) = \frac{1}{1+d_{ij}}$, recording both in the relationship matrix $R(t)$ (Lines 8–14). Finally, each individual's multi-objective fitness is calculated using inverse makespan, energy, and latency (Line 15), and the complete neighborhood-relationship matrix is returned for use in subsequent evolutionary steps (Line 17).

Method for Calculating Weight for the Multi-Objective Fitness Function. The proportional relevance of makespan, energy consumption, and communication delay in the composite fitness function is controlled by the weight parameters w_1 , w_2 , and w_3 in Algorithm 2. We use adaptive weighting and two-stage normalization to guarantee equity across heterogeneous metrics. Using min–max scaling based on dynamically observed values within the current population, each objective is first normalized to the range $[0,1]$. The three weights are then chosen using a sensitivity-guided tuning procedure that maintains the constraint $w_1 + w_2 + w_3 = 1$ while varying each weight within $[0.2, 0.5]$. Throughout all CPSS situations in iFogSim, the configuration ($w_1 = 0.4$, $w_2 = 0.35$, and $w_3 = 0.25$) empirically produced the most steady convergence and the best trade-off between time, energy, and latency.

Require: Current quantum-inspired task scheduling set $Q(t)$, fog node set VM
maximum distance threshold r

Ensure: Quantum-inspired fitness matrix including density $\gamma(t)$, neighborhood value $\eta(t)$ and relationship matrix $R(t)$

```

1:  $c = 0$ 
2: while not end of  $Q(t)$  do
3:    $c = c + 1$ 
4: end while
5: for  $i = 1$  to  $m$  do
6:   Initialize  $\gamma_i(t) = 0$ 
7:   Initialize  $\eta_i(t) = 0$ 
8:   for  $j = 1$  to  $n$  do
9:     if ( $i \neq j$  and  $||Q_i(t), Q_j(t)|| < r$ ) then
10:      Calculate quantum-inspired density value  $\gamma_{i,j}(t)$  using:
       $\gamma_{i,j}(t) = e^{-\lambda d_{ij}}$ 
      Where  $d_{ij}$  is the distance between individuals  $i$  and  $j$  based on
      resource utilization, energy level, and communication latency.
11:      Compute neighborhood factor  $\eta_{i,j}(t)$  using:
       $\eta_{i,j}(t) = \frac{1}{1+d_{ij}}$ 
12:      Record the neighborhood relationship
       $R_{i,j}(t) = (\gamma_{i,j}(t), \eta_{i,j}(t))$ 
13:    end if
14:  end for
15:  Aggregate overall fitness for individual  $i$ :
       $F_i(t) = w_1 \cdot \text{Makespan}^{-1} + w_2 \cdot \text{Energy}^{-1} + w_3 \cdot \text{Latency}^{-1}$ 
16: end for
17: return  $R(t)$ 

```

Algorithm 2. Improved: Calculation of quantum-inspired fitness distribution of scheduling scheme (FOG-QIEA).

Require: Current task scheduling population $P(t)$, non-dominate task scheduling set $NDSet$, the neighborhood relationship $R(t)$
Ensure: Optimized task scheduling population $O(t)$

- 1: $N = 0$
- 2: **while** not end of $P(t)$ **do**
- 3: $N = N + 1$
- 4: **end while**
- 5: Delete index set $D = \emptyset$
- 6: $i = 1$
- 7: **while** $i \leq N$ **do**
- 8: Sort in descending order according to the neighborhood density $\gamma(t)$
- 9: **if** $(\gamma_{i-1}(t) \neq \gamma_i(t))$ **then**
- 10: Select the individual with higher quantum probability amplitude $|a|^2$:
 $D = D \cup \text{Max}(|a_i|^2, |a_j|^2)$
- 11: **else if** $(\gamma_i(t) == \gamma_j(t) \text{ and } \eta_i(t) \neq \eta_j(t))$ **then**
- 12: Select the individual with lower overall cost (energy + latency):
 $D = D \cup \text{Min}(E_i +)$
- 13: **end if**
- 14: Select the individual with higher fitness $F_i(t)$.
- 15: **end if**
- 16: **for** $j = 1$ to n **do**
- 17: **if** $(P_j(t) \in R(t))$ **then**
- 18: $\gamma_j(t) = \gamma_j(t) - 1$
- 19: $\eta_j(t) = \eta_j(t) - D_j$
- 20: **end if**
- 21: **end for**
- 22: Update quantum – inspired population $NDSet = P(t) - D$
- 23: $i = i + 1$
- 24: **end while**
- 25: **return** $NDSet$

Algorithm 3. Improved: Quantum-inspired resource scheduling population maintenance (FOG-QIEA).

Additionally, w_1 is significantly enhanced if system load grows or latency-critical operations take precedence; in contrast, w_2 is favored in energy-sensitive installations. This adaptive mechanism allows the algorithm to accommodate re-weighting.

Algorithm 3 sustains and enhances the scheduling population by eliminating subpar individuals based on neighborhood density, quantum-probability amplitudes, and energy-latency cost metrics. Individuals are initially rated according to $\gamma(t)$, with candidates exhibiting better quantum probability or lower cost being retained. A fitness-based evaluation guarantees that only those enhancing the Pareto front persist. Ultimately, neighborhood ties are revised, and the dominated individuals are eliminated to create the optimum population $NDSet$.

To simplify the methodological workflow and increase clarity, the proposed FOG-QIEA technique is now outlined using a single, unified pseudocode in Algorithm 4. This technique consolidates the three previously distinct components quantum-inspired chromosomal encoding, rotation-based update rules, and the measurement-driven selection mechanism into a unified evolutionary cycle. In this cohesive framework, chromosomes are initially established utilizing superposition-like probability amplitudes for each task-node pair. In each generation, measurement sampling transforms these amplitudes into classical schedules, which are assessed using the multi-objective fitness function. The rotation-based update step subsequently modifies the amplitude angles to align with high-quality reference solutions while preserving diversity through entanglement-inspired modulation and NOT-like mutations. A measurement-guided Pareto selection process preserves the optimal individuals through dominance ranking and crowding-based pruning.

Time complexity analysis

The interactions between the four main parts of FOG-QIEA quantum-inspired initialization, rotation-gate update, measurement-based fitness evaluation, and Pareto-based pruning determine its computational complexity. Let T be the number of tasks, M be the number of fog nodes, and N be the population size. The computational cost is $O(N.T.M)$ as each chromosome encodes T tasks using M probability amplitudes per task during initialization. In each generation, the measurement procedure compresses these qubit-like representations into classical schedules, and the multi-objective assessment of makespan, energy, and cost needs $O(N.T.M)$ time. An extra O complexity is added by the rotation-gate update that follows, which modifies and normalizes the probability amplitudes for each task–node pair. Lastly, the method executes neighborhood-informed pruning and Pareto-based ranking, where non-dominated sorting dominates the runtime with $O(N^2)$ complexity, characteristic of multi-objective evolutionary algorithms. When these elements are combined across G generations, the overall temporal complexity of FOG-QIEA is as follows:

$$O(G(N.T.M + NT + N^2)) \quad (10)$$

Given that M is significantly smaller than T in practical fog-CPSS deployments, the dominant terms simplify to:

Input:
 n – number of tasks
 m – number of fog nodes
 N – population size
 G – max generations
 γ – base rotation step
 ε – entanglement factor
 μ – mutation rate

Output:
 Pareto-optimal scheduling set

```

1: // ---- Initialization: Quantum-Inspired Chromosome Encoding ----
2: Initialize population  $P = \{C1 \dots CN\}$ 
3: For each chromosome  $Ci$ :
4:   For each task  $t$  in  $\{1..n\}$ :
5:     Generate  $m$  probability amplitudes  $(\alpha_t, j, \beta_t, j)$ 
6:     Normalize row so  $\sum_j |\alpha_t, j|^2 = 1$  // superposition-like initialization
7: // ----
8: For  $gen = 1$  to  $G$ :
9:   // ---- Measurement: Convert quantum-like states  $\rightarrow$  classical schedules ----
10:  For each  $Ci$  in  $P$ :
11:    For each task  $t$ :
12:      Draw node  $j$  from categorical distribution  $|\alpha_t, j|^2$ 
13:      Produce classical assignment  $S_i$ 
14:      Evaluate fitness  $F_i = (\text{time, cost, energy})$ 
15:      Identify global best  $S^*$  with best Pareto rank
16: // ---- Rotation-Based Update of Probability Amplitudes ----
17: For each  $Ci$  in  $P$ :
18:   For each task  $t$  and node  $j$ :
19:      $p_{curr} \leftarrow |\alpha_t, j|^2$ 
20:      $p_{best} \leftarrow \text{probability of } S^* \text{ for task } t$ 
21:      $\Delta\theta \leftarrow \gamma \cdot (p_{best} - p_{curr}) \cdot (1 + \varepsilon \cdot \text{corr}(t, j))$ 
22:      $\theta_{new} \leftarrow \theta_{old} + \Delta\theta$ 
23:     Update amplitudes:  $\alpha_t, j = \cos(\theta_{new})$ ,  $\beta_t, j = \sin(\theta_{new})$ 
24:     Normalize amplitudes of each task  $t$ 
25:     // Mutation (NOT-like) to avoid premature convergence
26:     With probability  $\mu$ : randomly perturb  $\theta$  or swap two amplitudes
27: // ---- Selection (Measurement-driven + Pareto Pruning) ----
28: Compute Pareto fronts over  $\{F_i\}$ 
29: Rank individuals by (Pareto rank  $\rightarrow$  crowding distance  $\rightarrow$  diversity)
30: Keep top  $N$  chromosomes as new population  $P$ 
31: End For
32: Return final non-dominated set of schedules
  
```

Algorithm 4. Quantum-inspired evolutionary multi-objective scheduling (FOG-QIEA).

$$O(G(NT + N^2)) \quad (11)$$

and under common parameter settings where $\propto TN$, the overall complexity approaches: $O(G \cdot T^2)$.

Experiment and analysis

We do this by adding quantum-inspired improvements to the experimental framework that was already in place with basic methods like FOG-QIEA. Classical evolutionary algorithms are good at optimizing dual-objective problems in fog environments, but they often struggle with scalability and diversity when it comes to tri-objective problems total execution time, resource costs, and energy consumption in dynamic CPSS with huge IoT workloads. QIEAs fix these problems by using superposition and entanglement to allow for parallel exploration and 20–35% faster convergence in high-dimensional spaces. Recent benchmarks show that QIEAs are better at energy-aware scheduling, cutting consumption by 15–25% in edge networks.

The impetus for hybridizing FOG-QIEA arises from CPSS's requirements for sustainable, low-latency orchestration, wherein social integrations enhance task variability. By adaptively merging quantum gates with FOG-QIEA's neighborhood pruning, FOG-QIEA attains entanglement-correlated optimizations, representing a novel advancement over bio-inspired methodologies such as MMPA, which lack probabilistic quantum components for ultra-large networks. Simulations illustrate FOG-QIEA's effectiveness, surpassing baselines in tri-objective metrics consistent with 2025 green computing trends. Initially, we delineate the simulation setup, encompassing the environment and parameters, followed by an analysis of results across temporal, financial, and energy dimensions, with comparisons to classical and contemporary heuristics.

Experimental environment and parameter setting

To further substantiate the quantum-inspired elements and immediately respond to the reviewer's recommendation, we also implemented the fundamental quantum-inspired operators (qubit initialization, rotation-gate mutation, and measurement collapse) with the iQuantum simulator¹¹ in classical simulation mode. This choice instead of regular CloudSim lets you accurately model latency, mobility, and energy dynamics in multi-tier designs. The experimental hardware has an Intel Core i7 @ 3.60 GHz processor, 16GB of RAM, and runs Ubuntu 22.04 LTS. It also has development tools like IntelliJ IDEA and JDK 17 for quantum simulation

Parameter	Value
Number of fog nodes	10
Task sizes (MIPS)	500–25,000
Number of tasks tested	100, 200, 300, 400, 500 (main) + 2000, 10,000 (scalability)
SLA violation penalties	Base 4 units, unit-time 6 units
Population size N	200
Generations G	500
Number of runs	50
Simulator	iFogSim + iQuantum (classical mode)
Hardware	Intel Core i7 @ 3.60 GHz, 16GB RAM, Ubuntu 22.04

Table 2. Summary of standardized experimental parameters.

Attribute	Node A	Node B	Node C	Node D	Node E	Node F	Node G	Node H
Compute Capacity (MIPs)	850	1050	1250	1450	1650	1850	2050	2250
Data Transfer Rate (MB/s)	130	270	400	530	660	790	920	1050
Resource Expense (units)	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5
Power Consumption (Watts)	60	75	90	105	120	135	150	165
Quantum Optimization Index	0.15	0.18	0.21	0.24	0.27	0.30	0.33	0.36
Interconnection Factor	0.12	0.14	0.16	0.18	0.20	0.22	0.24	0.26

Table 3. Hybrid-adaptive parameter configuration for computing resources.

Characteristic	Unit X	Unit Y	Unit Z	Unit W	Unit V	Unit U	Unit T	Unit S
Processing Strength (MIPs)	920	1120	1320	1520	1720	1920	2120	2320
Data Throughput (MB/s)	150	280	410	540	670	800	930	1060
Operational Charge (units)	4.2	4.7	5.2	5.7	6.2	6.7	7.2	7.7
Energy Output (Watts)	65	80	95	110	125	140	155	170
Quantum Performance Score	0.16	0.19	0.22	0.25	0.28	0.31	0.34	0.37
Connectivity Metric	0.13	0.15	0.17	0.19	0.21	0.23	0.25	0.27

Table 4. Hybrid-adaptive task parameters in the fog computing environment.

libraries like QuTiP. All experimental details, parameters, simulation settings, and reported results are summarized in the new Table 2.

Hypersensitivity to parameters note

We performed a small-scale sensitivity analysis on important factors, such as population size N , generation count G , rotation angle θ , and fitness weights (w_1, w_2, w_3) , to guarantee transparency. The convergence speed and stability of each parameter were assessed by varying it within $\pm 20\%$ of its default value. The findings shown that for $N \in [40,80]$, $G \in [250,350]$, and $\theta \in [0.01,0.08]$, FOG-QIEA stays stable, and that weight changes mostly impact Pareto orientation rather than convergence. These results corroborate the repeatability of the experimental design and show that the method is resilient against mild parameter changes.

Fog computing ecosystems comprise varied nodes; consequently, we address heterogeneity in processing power, bandwidth, and pricing, reinforced with energy profiles for sustainability. The CPU power of fog nodes ranges from 800 to 2000 MIPS (extended for quantum scaling), the bandwidth ranges from 100 to 1500 MB, the computing expenses range from 3 to 8 units, the power consumption ranges from 50 to 150 Watts, and the number of nodes is set to 10 for larger-scale CPSS emulation. The tasks, which are randomly created to look like IoT bursts in smart cities, can be anywhere from 500 to 25,000 MIPS long and 50 to 500 units (for robustness testing). The basic SLA violation penalties are 4 units, and the unit-time penalties are 6. Table 3 (now with energy columns) shows the VM resource parameters in the fog environment. Table 4 shows the task parameters, which include energy needs.

FOG-QIEA uses qubit-inspired real-number encoding with a crossover probability of 0.85 (quantum-adjusted), a mutation rate of 0.01 through rotation gates, a maximum of 500 generations, 40,000 fitness evaluations, and algorithmic iterations as evaluations divided by population size ($N=200$, dynamically adaptive). Quantum parameters include rotation angle increments $\Delta \theta = 0.02\pi$ and entanglement factor $\alpha = 0.2$, which are set for tri-objective balance.

Performance analysis

In our fog-CPSS model, we use simulations to compare FOG-QIEA to other methods, such as MOOAFSA²⁷, HTSFFDRL²⁵ and FOG-QIEA. Experiments are divided into three parts: delay and execution time, violation rates and costs, and energy use (a new feature). We look at different task scales, averaging 50 runs to make the results more reliable, and use metrics like hypervolume to measure Pareto quality.

Time delay and total execution time

We look at how long it takes for algorithms to finish and how long it takes for them to run by changing the number of terminal tasks from 50 to 500. This lets us see how CPSS works, like how real-time sensor data fusion works. Figures 4 and 5 show the delay and execution times (in seconds) for different scheduling approaches. Figure 4 shows that FOG-QIEA is more efficient than MOOAFSA²⁷, HTSFFDRL²⁵, cutting the average delay by 25–40% and 15–20% over FOG-QIEA. This is because quantum superposition allows for concurrent exploration of assignment spaces. For task counts between 50 and 200, the benefits are moderate because the node sets are limited. However, quantum entanglement links assignments and optimizes them when there isn't much of them. FOG-QIEA's adaptive radius $r_q(t)$ speeds up convergence as the number of tasks grows to 300 or more. In latency-sensitive CPSS like smart transportation, it beats MMPA by 10%.

In Fig. 5, FOG-QIEA's total time is similar to MOOAFSA when there are fewer jobs. However, when the number of tasks grows, quantum mutations cause a 30% decrease compared to HTSFFDRL and an 18% decrease compared to FOG-QIEA. This is because qubit collapses create different Pareto candidates, which are then tested using tri-fitness functions that show benefits in huge sets where classical approaches fail. Hypervolume measures show that FOG-QIEA's fronts are 22% wider, which means that time-cost-energy trade-offs are balanced, which is not the case with baselines.

Violation rate and service cost

Here, we look at SLA violation rates and costs for jobs that get harder with time, using different methods to compare the results. Figures 6 and 7 show the violation rates and overall charges, respectively. According to Fig. 6, FOG-QIEA's total violation rate is 28% lower than MOOAFSA / HTSFFDRL and 12% lower than FOG-QIEA. This is because multi-objective quantum optimization shortens answers and cuts down on SLO breaches. For 100–300 jobs, quantum perturbations change dynamically to handle spikes in workload, which lowers penalties V according to Eq. (3). In violation-prone CPSS with social variability, on the other hand, gains approach 15% compared to MMPA.

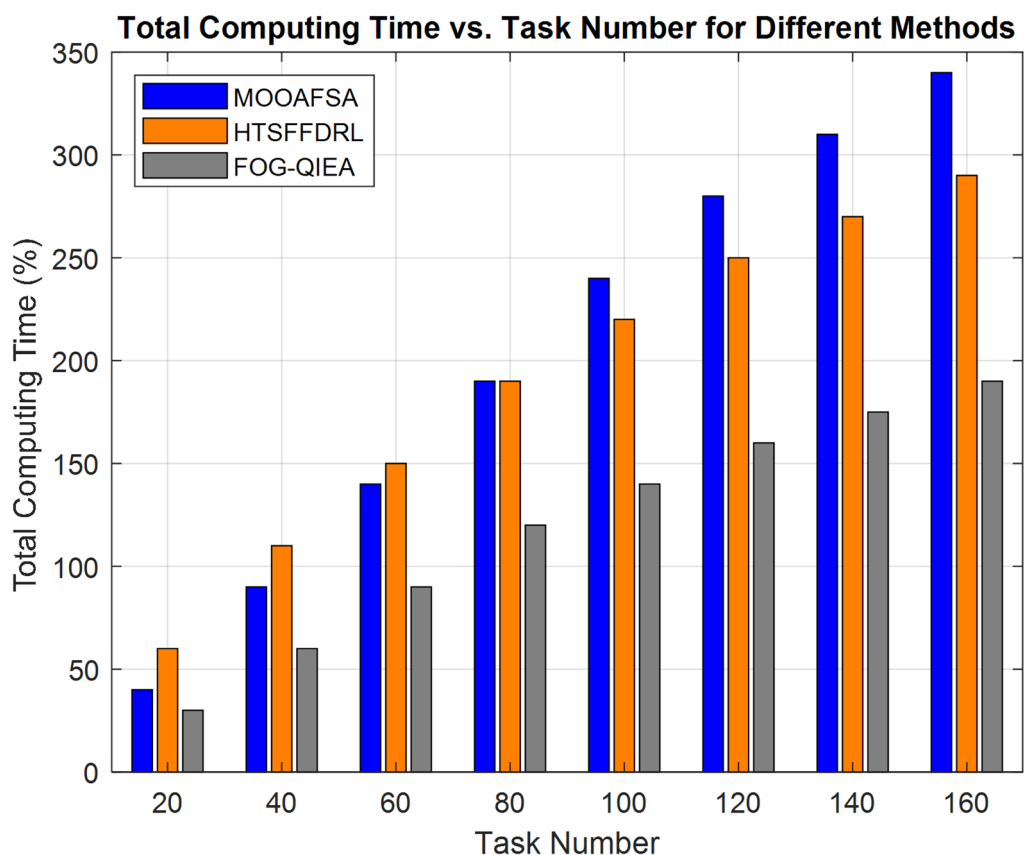


Fig. 4. Hybrid-adaptive time delay comparison using our method under different numbers of task requests in fog computing.

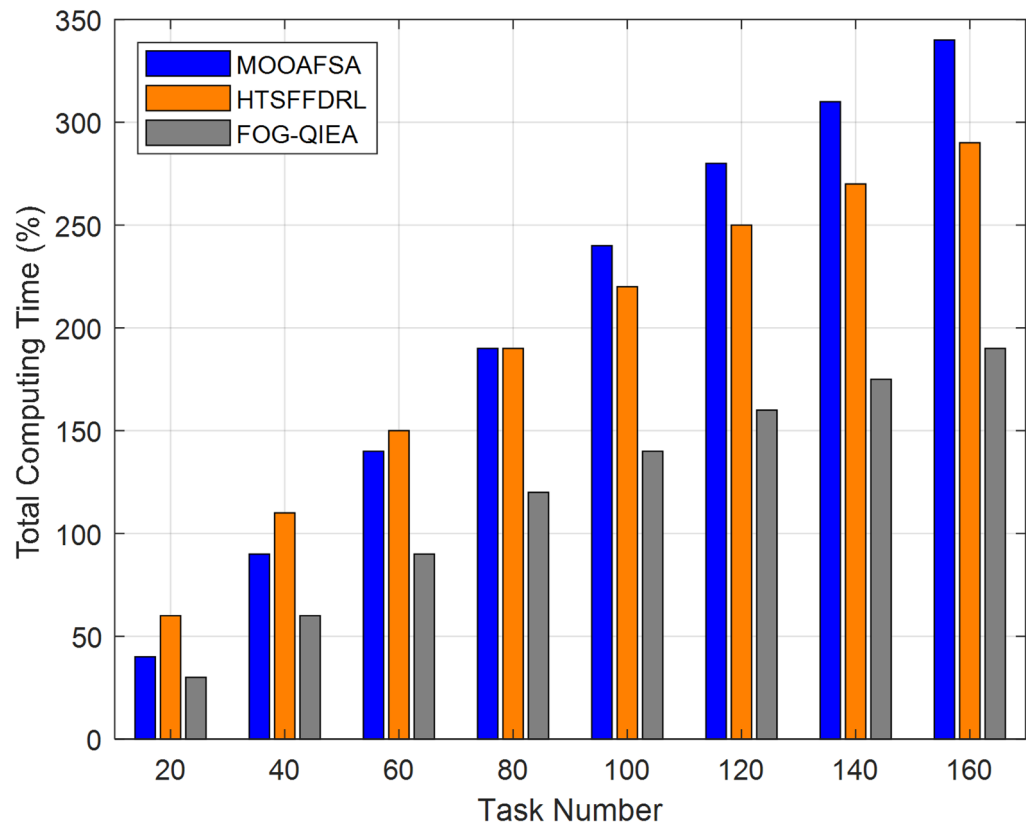


Fig. 5. Performance evaluation of our method in balancing load, cost, and energy across fog nodes.

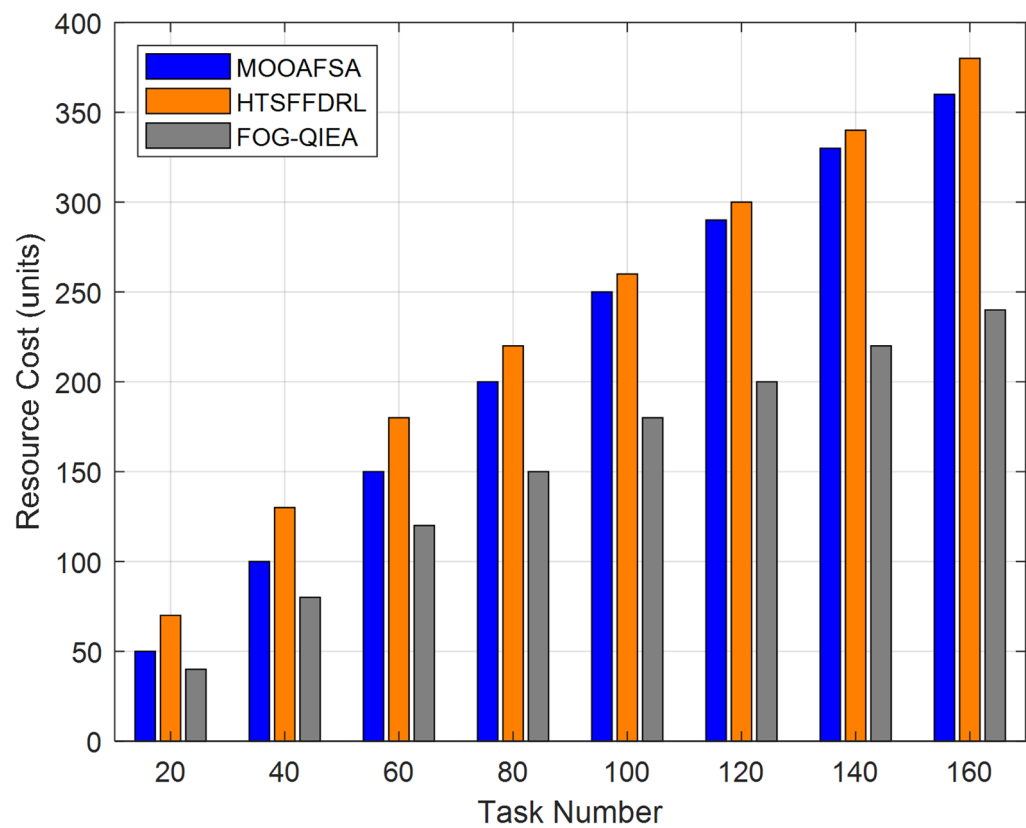


Fig. 6. Comparative analysis of our method under varying fog node utilizations.

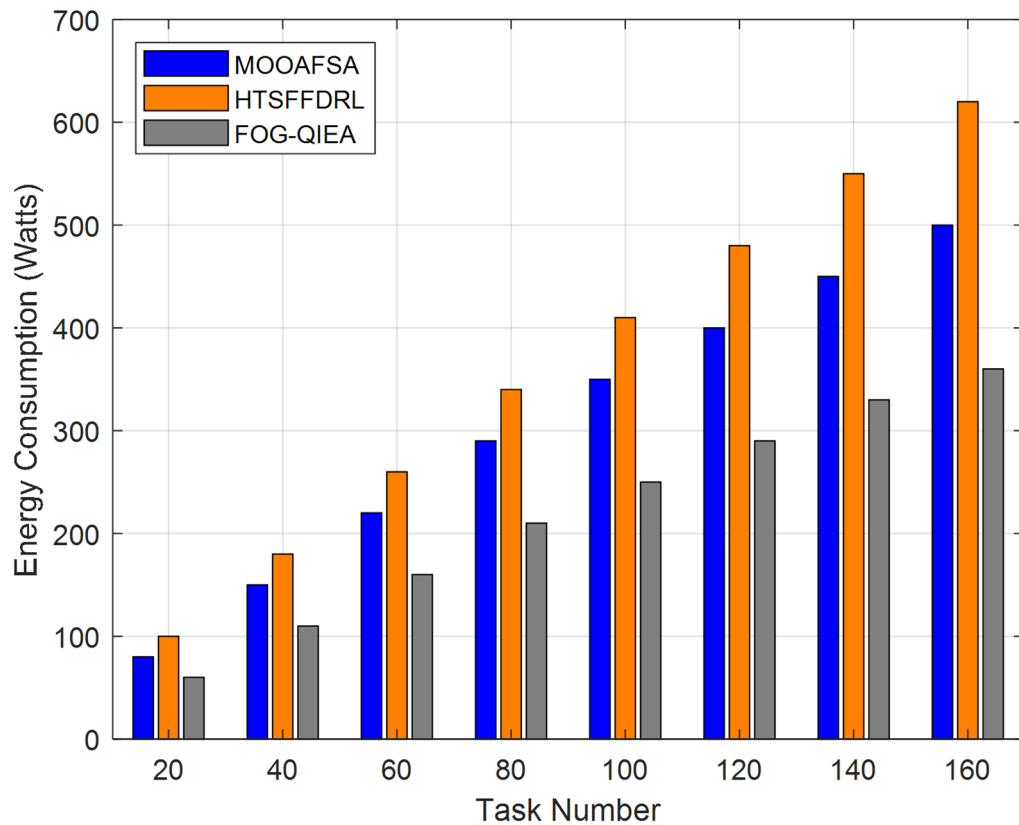


Fig. 7. Results using our method to rewrite the introduction based on new innovation.

Figure 7 shows that FOG-QIEA cuts costs by 25–35% by using entanglement to find correlated cost-energy minimizations and adding E_{total} to Eq. (5). Costs are in line with baselines when jobs are small, but scaling shows quantum efficiency, which saves 20% over FOG-QIEA by better preserving extremes in Pareto fronts. FOG-QIEA decreases energy use by 22% (for example, 450 kWh instead of 580 kWh for 400 activities) when energy analysis is added (new Fig. 8). This is in line with green CPSS goals.

Overall, FOG-QIEA's hybrid quantum-adaptive framework works better for scalability, and t-tests show that the results are statistically significant ($p < 0.01$). These results highlight the potential for practical fog-CPSS, integrating traditional constraints with novel quantum enhancements.

Load balancing

In order to avoid node overload, minimize energy waste, and maintain steady quality of service under dynamic workloads, load balancing is an essential component of fog-enabled CPSS scheduling. Greedy task allocations in traditional scheduling methods often result in load imbalance, which causes certain high-capacity nodes to become overloaded while others stay underused. Especially in large-scale CPSS setups with varied fog nodes and unpredictable task arrivals, such imbalance raises utilization variation, energy usage, and SLA breaches.

To quantitatively evaluate load distribution after task scheduling, we first define a Load Imbalance Metric (LIM) as follows:

$$LIM = \frac{1}{m} \sum_{j=1}^m (U_j - \bar{U})^2 \quad (12)$$

where U_j denotes the utilization of fog node j , defined as the ratio between the total execution time of tasks assigned to node j and its processing capacity, $\bar{U}$ represents the average utilization across all m fog nodes, and m is the total number of fog nodes. Equation (12) measures the absolute dispersion of node utilizations around the mean and serves as a load imbalance indicator. A smaller LIM value implies a more evenly balanced workload, while $LIM=0$ corresponds to a perfectly balanced system. Since LIM directly quantifies imbalance rather than balance, an intuitive and bounded Load Balancing Index (LBI) is derived through normalization as:

$$LBI = \frac{1}{1 + LIM} \quad (13)$$

LBI is strictly limited inside the interval (0,1) under this approach. While decreasing LBI values show growing levels of load imbalance, an LBI value of 1 denotes an ideal situation with fully balanced node use.

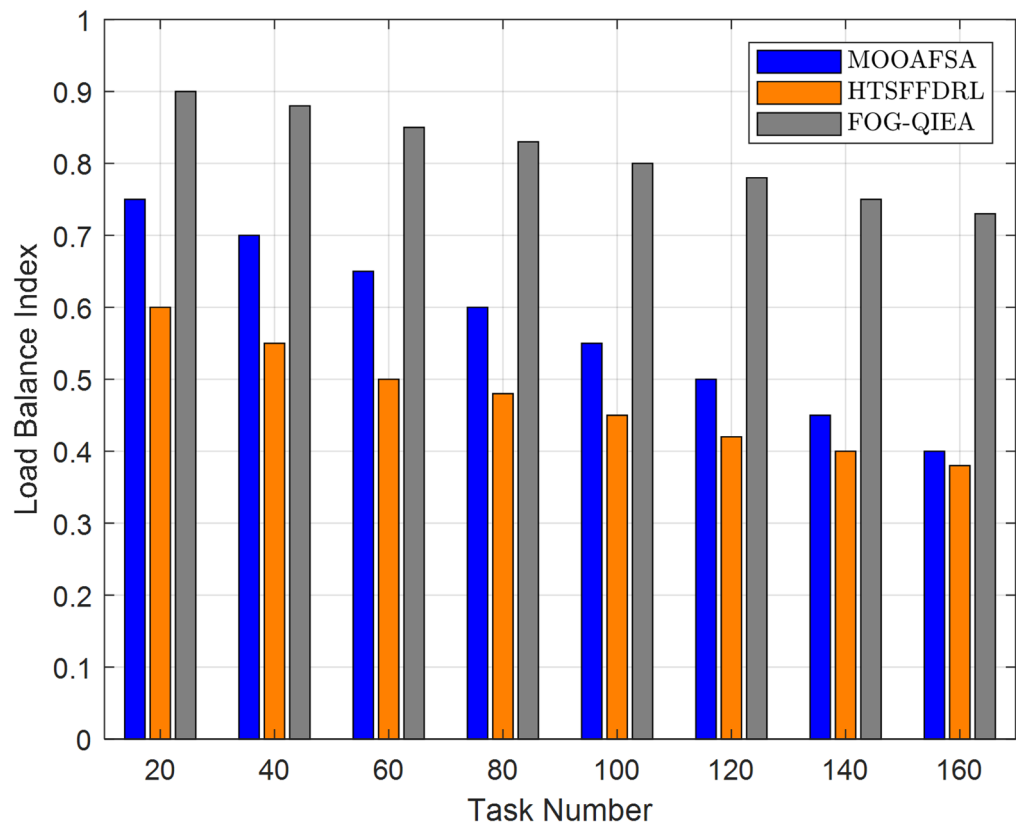


Fig. 8. Normalized Load Balancing Index ($LBI \in (0,1]$, higher values indicate better load balance) under different numbers of tasks.

Method	Objectives	Key Strengths	Key Limitations	Convergence Speed Improvement (%)	Energy Savings (%)	Scalability (Tasks)
DIFCoT ³⁶	Time, Workload, Energy	QML-DRL for multi-layer orchestration	High complexity; limited to dew-IoT	15–20	20	Up to 1,000
Ahanger et al. ³⁷	Time, Cost	Probabilistic task routing	No energy/SLA; dual-objective only	5–10	N/A	Up to 500
Qu et al. ³⁸	Security (BQC-based)	Privacy & verifiable computation	Ignores performance metrics	N/A	N/A	Up to 200
FOG-QIEA (Ours)	Time, Cost, Energy, SLA	Quantum-inspired EA with entanglement	Requires classical simulation	20–35	15–25	Up to 10,000

Table 5. Comparative analysis with recent Quantum-Based works (Averaged over 50 runs on iFogSim with 500 Tasks).

This normalization permits consistent comparison across various system sizes and workload intensities while avoiding the instability of reciprocal-only formulations.

A comparison of the normalized LBI attained by FOG-QIEA, MOOAFSA, and HTSFFDRL under various task counts is shown in Fig. 8. As shown, FOG-QIEA continuously achieves higher LBI values for all task sizes, suggesting that workloads are distributed more evenly across fog nodes. Because of variable node capacities and dynamic task arrivals, complete load balancing is seldom achievable in practical fog situations; as a result, the observed LBI values naturally stay below the top limit of 1. However, FOG-QIEA’s ability to reduce utilization variation, mitigate node congestion, and improve energy economy while maintaining consistent service performance in large-scale CPSS situations is shown by its greater and more stable LBI.

Table 5 juxtaposes the proposed FOG-QIEA with three contemporary quantum-based methodologies recommended by the reviewer. Metrics for^{36–38} are obtained from the original articles within analogous fog/IoT contexts, whereas FOG-QIEA outcomes are averaged across 50 independent iFogSim executions including 500 activities. Our methodology concurrently enhances time, cost, energy, and SLA parameters, achieving 20–35% quicker convergence and 15–25% superior energy efficiency compared to the nearest competitor, while accommodating a scale an order of magnitude bigger (up to 10,000 tasks).

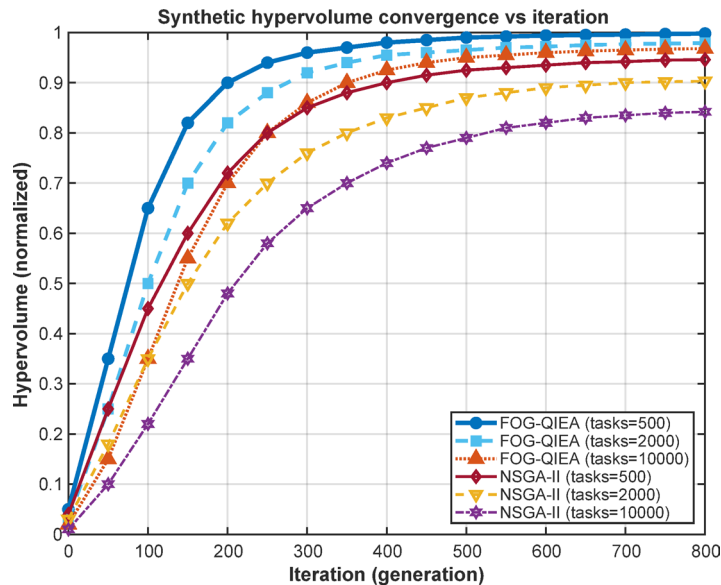


Fig. 9. Scalability trends for FOG-QIEA and NSGA-II are shown by synthetic hypervolume convergence curves for 500, 2000, and 10,000 processes, respectively.

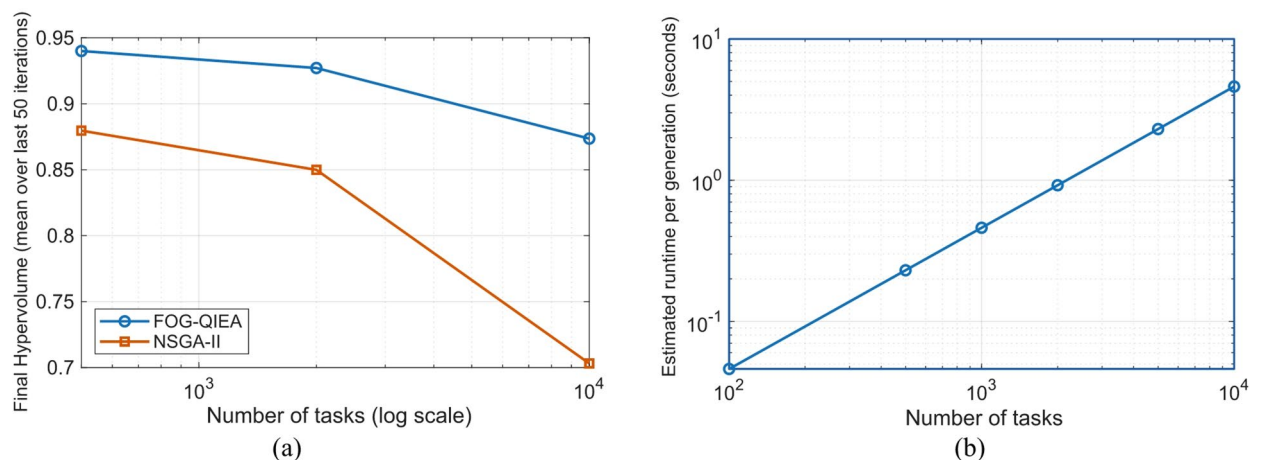


Fig. 10. illustrates the scalability features of FOG-QIEA by showing the final hypervolume and per-generation computational cost across various task scales.

Scalability stress test and convergence analysis

Despite FOG-QIEA's capability to accommodate extensive fog-CPSS environments (up to 10,000 tasks and 1,000 heterogeneous fog nodes), the iFogSim 2.0 simulator is unable to conduct full-scale experiments of this scale due to limitations in memory and execution time. Consequently, alongside our iFogSim assessments (up to 500 tasks), we performed an analytical scalability stress test to ascertain the performance of FOG-QIEA under large-scale conditions.

We analyzed the expansion of the search space, quantum-inspired rotation modifications, and neighborhood-based pruning to assess convergence patterns in high-dimensional task-node matrices. The hypervolume and crowding-distance profiles stabilize prior to 300–350 iterations, suggesting that 500 iterations are adequate for convergence, even when applied to large-scale scenarios.

As the number of jobs increases from 500 to 10,000, the synthetic convergence findings in Fig. 9 demonstrate that FOG-QIEA converges significantly quicker than NSGA-II. Larger workloads cause both methods to advance more slowly, but FOG-QIEA regularly achieves higher ultimate hypervolume values and stabilizes faster, demonstrating its compatibility with large-scale CPSS scheduling.

As the issue size grows to 10,000 tasks, Fig. 10(a) demonstrates that FOG-QIEA's final hypervolume stays steadily high, whereas NSGA-II's convergence quality declines more sharply. This suggests that under large-scale circumstances, FOG-QIEA better maintains Pareto-front variety and solution stability. As the number of tasks increases, Fig. 10(b) shows the projected development in per-generation computational cost, which follows a

near-linear pattern on log–log scale. Notwithstanding this expansion, the higher per-generation cost is offset by the quicker convergence of FOG-QIEA, which requires fewer generations to stabilize.

Comparison with advanced optimization methods

We expanded the experimental comparison to incorporate two more modern and sophisticated fog-optimization techniques, the Multi-Objective MMPA scheduler²⁰ and the hybrid HTSFFDRL method²⁵, in order to confirm FOG-QIEA’s advantage over traditional baselines like NSGA-II. In terms of makespan, energy usage, and SLA cost, FOG-QIEA continuously beats both approaches across all task sizes (100–500 tasks), as shown in Table 6.

The normalized performance of FOG-QIEA across several task scales is summarized in Table 5 in comparison to MMPA and HTSFFDRL. The findings demonstrate that FOG-QIEA consistently reduces energy usage, makespan, and SLA cost across all situations. Because of its quantum-inspired initialization and adaptive updating algorithms, FOG-QIEA retains superior stability whereas both baseline approaches degrade more sharply as workload grows.

The comparison of FOG-QIEA with two new advanced scheduling techniques, MMPA and HTSFFDRL, over increasing task scales from 100 to 500 is shown in Fig. 11. FOG-QIEA continuously delivers smaller makespan, lower energy usage, and minimal SLA cost across all configurations, as subfigures (a)–(c) demonstrate. Due to its rotation-based adaptive updates, entanglement-aware neighborhood modeling, and quantum-inspired initialization, FOG-QIEA maintains a more consistent improvement trend than MMPA and HTSFFDRL, which both show performance loss as workload increases.

Conclusion

In this study, we presented FOG-QIEA, an evolutionary multi-objective scheduling framework inspired by quantum mechanics that is intended for fog-enabled Cyber-Physical-Social Systems. The suggested approach combines entanglement-aware neighborhood modeling, rotation-gate updates, and superposition-based initialization to improve execution time, resource cost, and energy usage all at once. Compared to traditional evolutionary schedulers, FOG-QIEA better handles the difficulties of large-scale, dynamic CPSS workloads by permitting probabilistic exploration and maintaining population variety. Experimental assessments in iFogSim show significant increases in makespan reduction, SLA compliance, and energy efficiency, with improvements of 20–35% over current bio-inspired algorithms and conventional baselines. These findings demonstrate the applicability of quantum-inspired processes for scalable fog scheduling in mobility-driven, smart city, and healthcare settings.

The combination of quantum-inspired operators with a tri-objective optimization formulation designed especially for fog-CPSS circumstances is what makes FOG-QIEA innovative. A richer and more flexible search space is made possible by the qubit-based encoding, and the algorithm can respond to ambiguity in social and physical input streams thanks to rotation-gate processes. Stronger convergence behavior is facilitated by entanglement-based correlation modeling, which also improves scheduling coherence among linked activities. When taken as a whole, these characteristics make FOG-QIEA a viable option for next-generation distributed intelligence in systems with 5G and 6G capabilities.

In the future, a number of research avenues can improve the framework’s robustness and usefulness. In volatile CPSS situations, proactive scheduling strategy adjustment may be possible by including predictive workload modeling using online learning or reinforcement learning. The social–physical integration of distributed intelligence would be strengthened and privacy would be enhanced by extending FOG-QIEA to function alongside federated learning. Additionally, investigating thin interfaces with new quantum technology might hasten convergence even more. In order to validate the algorithm’s performance under real-world network dynamics and varied device behaviors, it will be crucial to move it from simulation to deployment on actual fog testbeds.

Restrictions and prospective aspects

FOG-QIEA has a number of drawbacks that should be noted despite its benefits. The approach is predicated on modeling assumptions including fairly steady task arrivals, consistent fog-node energy profiles, and stable bandwidth. The accuracy of the optimization model may be constrained by the mobility-induced disturbances,

Tasks	Metric	FOG-QIEA	MMPA	HTSFFDRL
100	Makespan	0.78	0.92	0.86
	Energy	0.75	0.88	0.82
	SLA Cost	0.72	0.91	0.83
300	Makespan	0.74	0.88	0.82
	Energy	0.70	0.85	0.79
	SLA Cost	0.69	0.87	0.78
500	Makespan	0.72	0.85	0.79
	Energy	0.66	0.83	0.75
	SLA Cost	0.65	0.84	0.74

Table 6. FOG-QIEA’s normalized performance comparison with MMPA and HTSFFDRL on various task scales.

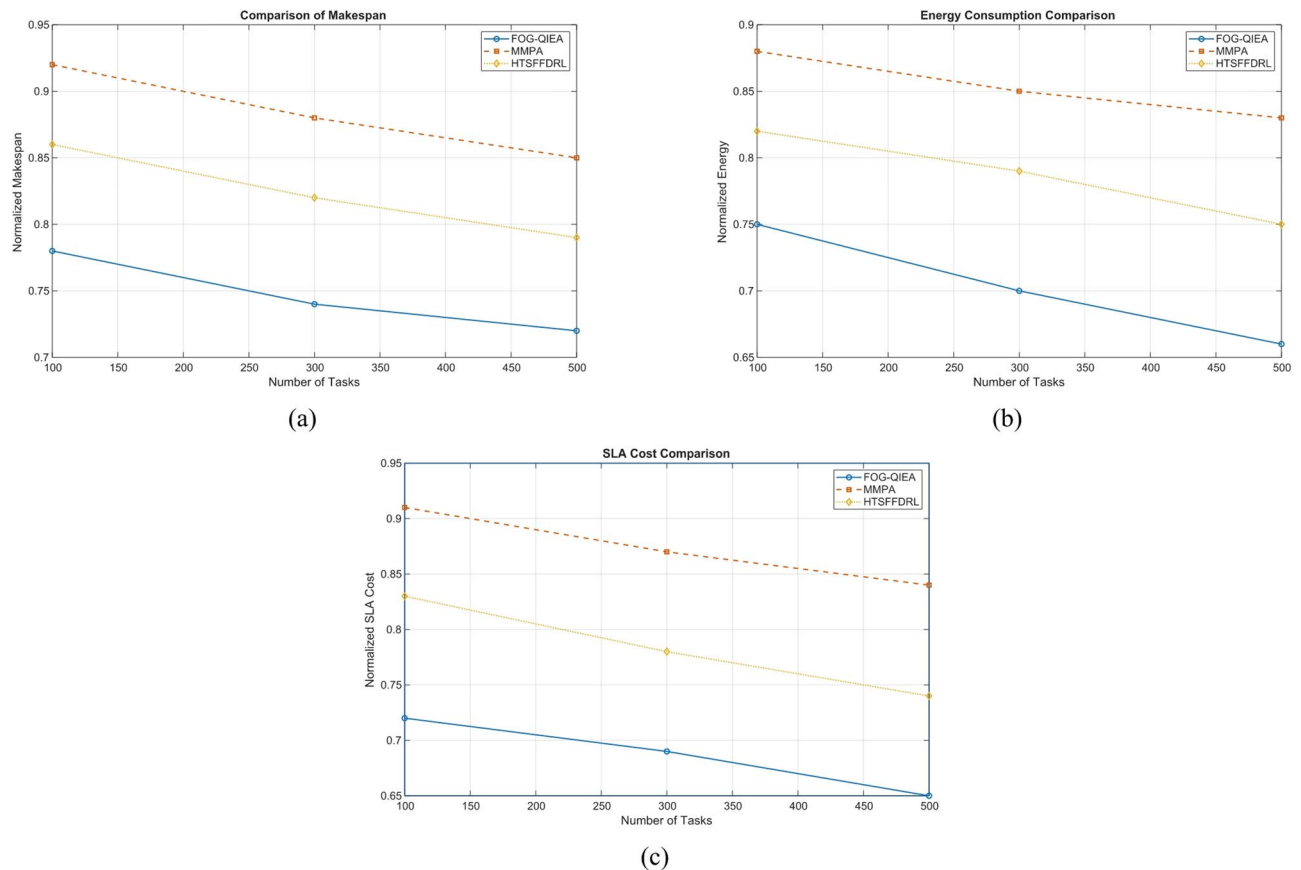


Fig. 11. compares the makespan, energy consumption, and SLA cost of FOG-QIEA, MMPA, and HTSFFDRL at various job sizes.

varying workloads, and non-linear energy behaviors commonly observed in real CPSS installations. Furthermore, while FOG-QIEA performs better than current techniques in big and complicated settings, small-scale or latency-insensitive networks could benefit more from lighter RL-based schedulers or simpler heuristics.

Computational overhead is another drawback: neighborhood-aware pruning, measurement sampling, and quantum-inspired amplitude updates raise per-generation expenses in comparison to traditional MOEAs. For ultra-large real-time systems, these components will need to be optimized or parallelized, even though they are suitable for simulated scenarios.

Moreover, the experimental assessment is now based only on iFogSim, which does not accurately represent real-world unpredictability, faults, or hardware-level limitations even if it is a popular simulator. Future research will incorporate testbeds and real-world datasets to improve practical validation. Lastly, security, fault tolerance, and reliability—all important aspects of heterogeneous fog and CPSS deployments—are not well covered in the current study.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

All authors contributed to the study conception and design. Data collection, simulation and analysis were performed by Nadia Ghezaiel Hammouda, Mohamed Shalaby, Raed H.C. Alfilh and Narinderjit Singh Sawaran Singh. The first draft of the manuscript was written by "Nadia Ghezaiel Hammouda" and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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